Pharmacology of
Drugs Used in Coagulation Disorders, Part 2
Fibrinolytics, Antiplatelet Agents, and Antidotes for the Anticoagulants

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Make.
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Following this lecture, students should be able to:

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1. List the pharmacokinetic properties of each class of drugs affecting blood coagulation and give specific examples.
2. Explain the structure-activity relationship of each class of the classes and give specific examples.
3. Relate the mechanisms of action of each class of drugs to their effects on blood coagulation and give specific examples.
4. Explain the clinical use of each class of drugs and give specific examples.
5. Describe the mechanisms of toxicities, drug-disease interactions and drug-drug interactions, giving specific examples.

Preparation Materials (links are in the CPG and on the next slide)

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Required

- ScholarRx Bricks | Practice Questions | See next slide.

Highly relevant optional materials:

- Dr. Goldstein's Word handout | Video Lecture | Guided Reading Questions
- Textbook resources with links are listed on the next slide

SUGGESTIONS:

- *Use the resources that work best for you.*
- *You do not need to study all of them.*
- *Work through the GUIDED READING QUESTIONS with pen/pencil and paper.*

Try answering them in your OWN WORDS first without looking at the readings, and then again after reviewing.

- *Practice questions (not graded): Simple Recall and Case Vignettes*

Links to resources

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Scholar Rx Bricks:
Name of Discipline/ Organ System: Hematology > Hemostasis Disorders > Thrombotic Disorders
Title of Brick: Anticoagulant drugs
Link: https://exchange.scholarrx.com/brick/anticoagulant-drugs

Access Medicine Katzung & Vanderah's Basic & Clinical Pharmacology, 16e, 2024; Chapter 34: Agents Used in Disorders of Coagulation
https://accessmedicine-mhmedical-com.nyit.idm.oclc.org/content.aspx?bookid=3382&sectionid=281752540

LWW Health Libraries, Premium Basic Sciences: Lippincott's Illustrated Reviews, Pharmacology, 8e, 2023; Chapter 13: Anticoagulants and Antiplatelet Agents
https://premiumbasicsciences-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=253326640&bookid=3222

Access Medicine Katzung & Trevor's Pharmacology Examination and Board Review, 13e, 2021; Chapter 34: Agents Used in Disorders of Coagulation
https://nyit.idm.oclc.org/login?url=https://accessmedicine-mhmedical-com/content.aspx?bookid=3058&sectionid=255306288

Key Concepts: Fibrinolytics

- Fibrinolytics alteplase, tenecteplase, and reteplase are recombinant tPA (tissue plasminogen activator) that convert fibrin-bound plasminogen to plasmin, which cleaves fibrin, which lyses thrombi. Fibrinolytics are given early in clot formation because clots become more resistant to fibrinolysis as they age.
- Fibrinolytics are administered IV (usually) for treatment of ischemic stroke (within 4.5 hours), acute myocardial infarction if PCI is unavailable, and treatment of PE. A fibrinolytic may be used to restore catheter and shunt function.
- Hemorrhage is a major adverse effect. A systemic lytic state can result from plasmin generation and degradation of other coagulation factors.
- Contraindications include pregnancy and patients with healing wounds, history of stroke, brain tumor, head trauma, intracranial bleed, and metastatic cancer.
- Streptokinase is a protein produced by beta-hemolytic streptococci. It forms a complex with plasminogen. The complex induces a conformational change in plasminogen that exposes its active site. Plasminogen then converts free and clot-bound plasminogen molecules to plasmin. Streptokinase produces a systemic lytic state. It is no longer available in the US.

Key concepts: Antiplatelet agents

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- Platelet aggregation inhibitors decrease the formation of a platelet-rich clot or decrease the action of chemical signals that promote platelet aggregation.
- The antiplatelet agents in this lecture block platelet COX-1 synthesis of thromboxane (aspirin), inhibit purinergic $P2Y_{12}$ ADP receptor (clopidogel, prasugrel, ticagrelor, cangrelor), and block GP IIb/IIIa receptors (eptifibatide, tirofiban).
- Low-dose aspirin is used for primary and secondary prevention myocardial infarction transient cerebral ischemia. It may be used with another antiplatelet agent
- $P2Y_{12}$ receptor blockers are used for prevention of thrombotic events in patients with recent MI or stroke, percutaneous coronary intervention (PCI), acute coronary syndrome, and established peripheral artery disease.
- GP IIb/IIIa inhibitors are intravenous adjuncts to PCI.
- As with all drugs that interfere with platelet function or coagulation, bleeding is the most important adverse effect of this class of drugs.

Key concepts: Reversal Agents

- Protamine sulfate is a highly cationic protein that combines and forms a stable complex with heparin and neutralizes it.
- Vitamin K (phytonadione) reverses warfarin anticoagulation by outcompeting warfarin at VKORC1. It is administered orally or IV for severe bleeding.
- Andexanet-alfa is inactivated factor Xa for reversal of anticoagulation by the oral direct factor Xa inhibitors (DOACs) rivaroxaban and apixaban.
- Idarucizumab binds specifically to dabigatran and its active acylglucuronide memtabolites, reversing the anticoagulant activity.
- 4-factor PCC from human plasma consists of unactivated factors II, VII, XI and X, Protein C, and Protein S. It is administered for urgent reversal of bleeding due to acquired coagulation factor deficiency induced by vitamin K antagonists and DOACs.

Fibrinolytic Agents

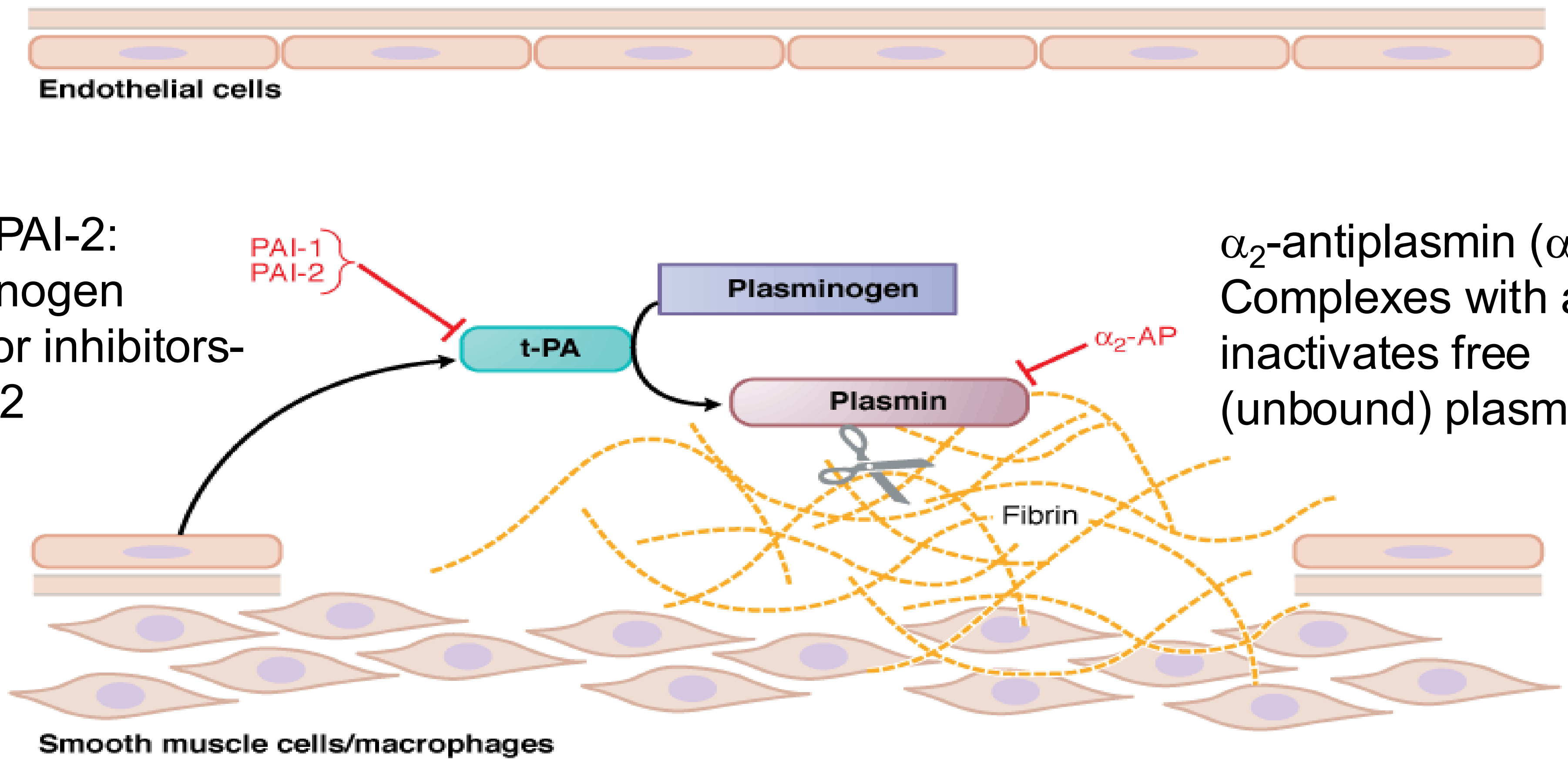
✳Alteplase
Tenecteplase
Reteplase

Tissue plasminogen activator (tPA) and derivatives

Streptokinase (natural product)

Fibrinolysis: The process of fibrin digestion by the fibrin-specific protease, plasmin.

PAI-1, PAI-2: Plasminogen activator inhibitors-1 and -2



α_2-antiplasmin (α_2-AP): Complexes with and inactivates free (unbound) plasmin

Source: Brunton LL, Chabner BA, Knollmann BC: *Goodman & Gilman's The Pharmacological Basis of Therapeutics, 12th Edition:* www.accessmedicine.com

In response to injury, endothelial cells synthesize and release tPA.

- tPA binds fibrin and activates ***fibrin-bound plasminogen*** several 100-fold more rapidly than it activates plasminogen in the circulation.
- Plasmin is the active fibrinolytic enzyme that remodels the thrombus and limits its extension by proteolytic digestion of fibrin.

Structure-activity relationships of t-PA fibrinolytics

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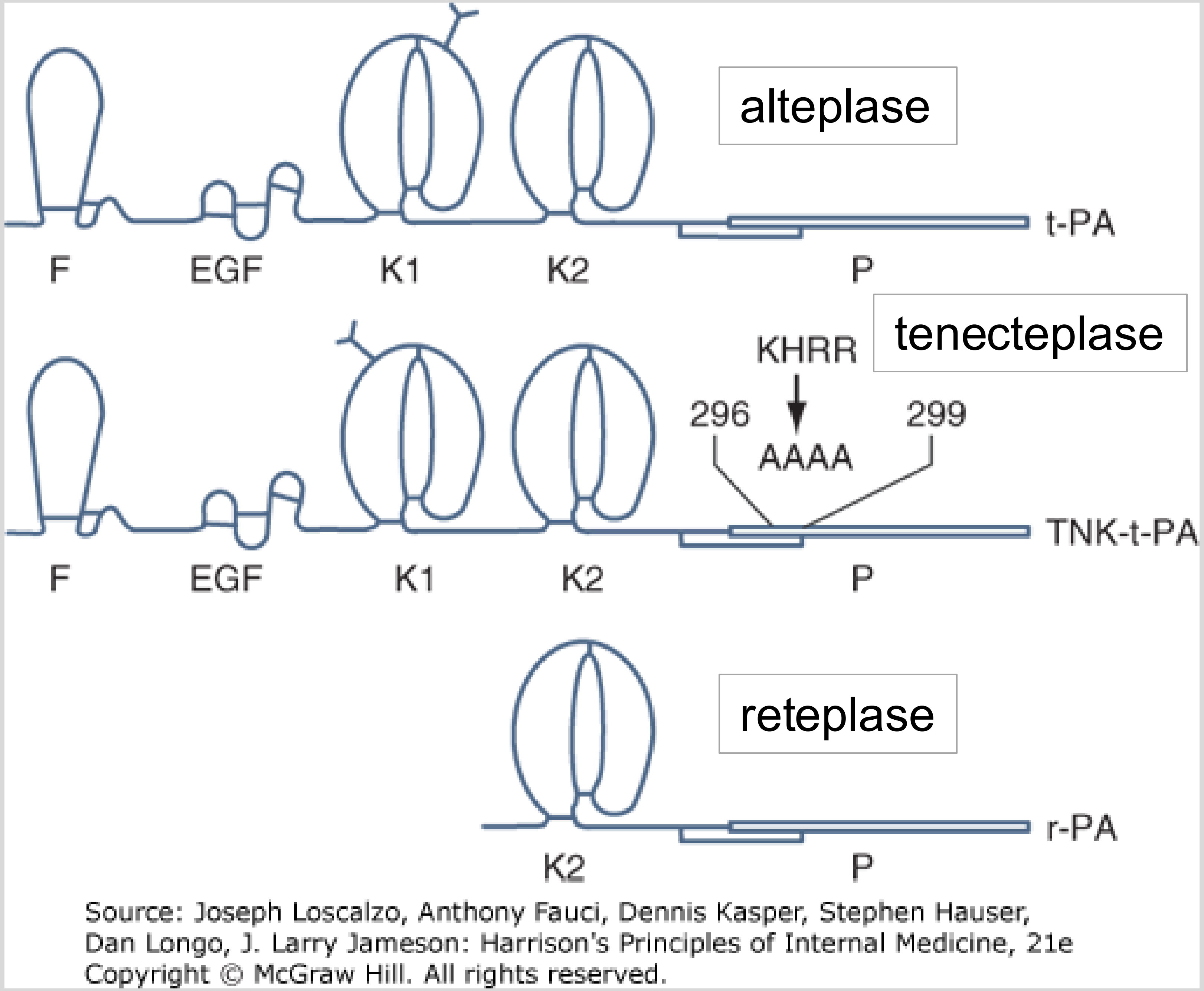


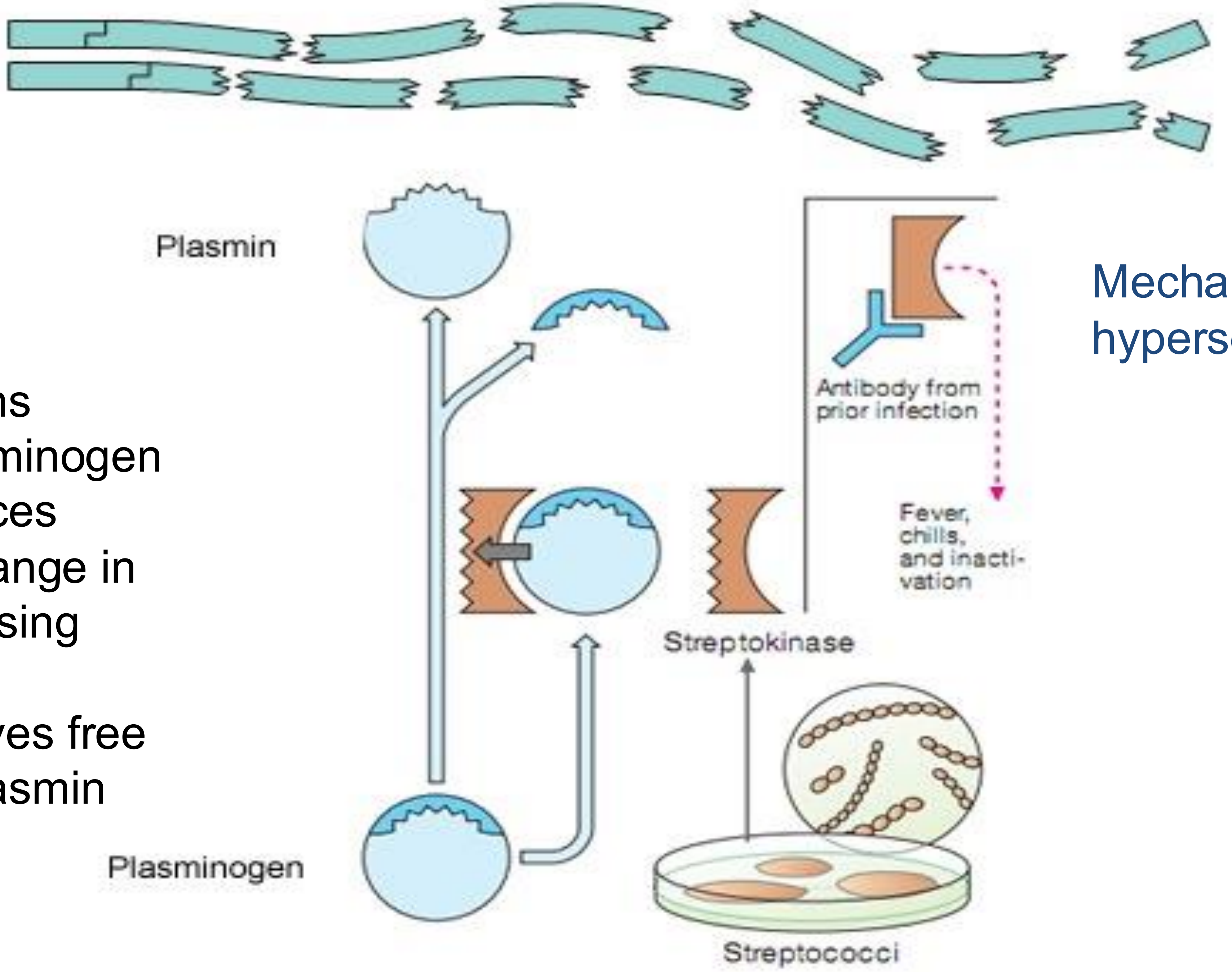
Harrison's, 21e; Figure 118-9. F:finger; EGF: epidermal growth factor
K1, K2: first and second kringle domains, respectively); P: protease domain

- Alteplase (t-PA)
  Recombinant version of natural protein
- Tenecteplase:
  1. K1glycosylation site (Y) repositioned → ↑ t½
  2. Protease domain substitution → resistance to PAI-1 inhibition
- Reteplase:
  Nonglycosylated, truncated variant
  Longer t ½ than alteplase
  Binds fibrin more weakly

Pharmacology of Fibrinolytic Agents: tPA and Derivatives

PK	IV, rapid hepatic clearance Alteplase t½ <5 min | Tenecteplase t½ ~20 min, $t½_{terminal}$ 90-130 min | Reteplase t½ ~15 min
MOA	Fibrinolytic agents catalyze the formation of plasmin from its precursor, plasminogen. The drugs induce rapid dissolution of thrombi. Both protective hemostatic thrombi and target thromboemboli are broken down, which can cause a generalized lytic state → hemorrhage
Uses	1- Acute ischemic stroke to ↓ neuronal death / infarction admin within 4.5 h 2- STEMI if PCI cannot be performed within 120 min, admin within 30 min of first medical contact; may still consider in patients who present late and have ongoing ischemia, admin within 12-24 h of symptom onset (with antiplatelet and anticoagulant) 3- PE treatment 4- Central venous catheter / dialysis catheter clearance (alteplase)
AE	Hemorrhage / hemorrhagic stroke, the major toxicity, resulting from lysis of hemostatic plugs at site of injury and from systemic plasmin generation, which degrades coagulation factors (esp. V and VII) as well as fibrin.

Streptokinase is a protein produced by β-hemolytic streptococci.



Mechanism of action

- Streptokinase forms complex with plasminogen
- The complex induces conformational change in plasminogen exposing active site
- The complex cleaves free plasminogen to plasmin

Mechanism of hypersensitivity reaction

Streptokinase binds both fibrin-bound and circulating plasminogen and it produces a systemic lytic state

Absolute and Relative Contraindications to Fibrinolytic Therapy

ABSOLUTE CONTRAINDICATIONS	RELATIVE CONTRAINDICATIONS
Prior intracranial hemorrhage	Uncontrolled hypertension (SBP >180 mm Hg or DBP >110 mm Hg)
Known structural cerebral vascular lesion	Traumatic or prolonged CPR or major surgery within 3 weeks
Known malignant intracranial neoplasm	Recent (within 2-4 weeks) internal bleeding
Ischemic stroke within 3 months	Noncompressible vascular punctures
Suspected aortic dissection	For streptokinase: prior exposure (>5 days ago) or prior allergic reaction to streptokinase
Active bleeding or bleeding diathesis (excluding menses)	Pregnancy
Significant closed-head trauma or facial trauma within 3 months	Active peptic ulcer
Goodman & Gilman 13e Table 32-3 CPR, cardiopulmonary resuscitation INR, international normalized ratio	Current use of warfarin and INR >1.7

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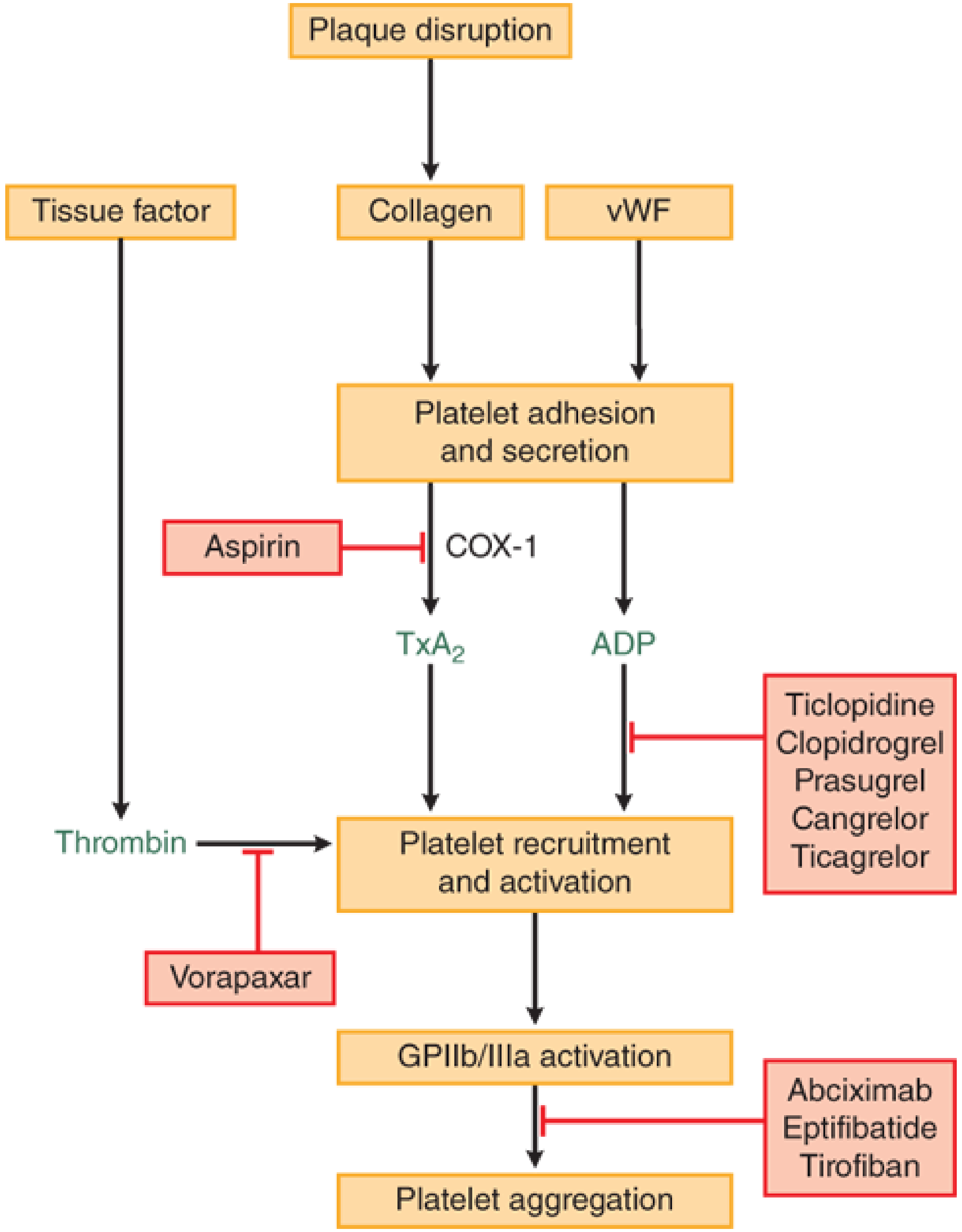
Antiplatelet Agents

Aspirin

ADP Receptor Antagonists

Glycoprotein IIb/IIIa Inhibitors



Antiplatelet agents decrease platelet aggregation for the prevention of cardiovascular adverse events.

Sites of action of antiplatelet drugs:

- Aspirin inhibits TxA_2 synthesis by irreversibly acetylating COX-1. Reduced TxA_2 release attenuates platelet activation and recruitment to the site of vascular injury.
- Ticlopidine, clopidogrel, and prasugrel irreversibly block $P2Y_{12}$, a key ADP receptor on the platelet surface.
- Cangrelor and ticagrelor are reversible inhibitors of $P2Y_{12}$.
- Abciximab, eptifibatide, and tirofiban inhibit the final common pathway of platelet aggregation by blocking fibrinogen and von Willebrand factor (vWF) from binding to activated GPIIb/IIIa.
- Vorapaxar inhibits thrombin-mediated platelet activation by targeting protease-activated receptor-1 (PAR-1), the major thrombin receptor on platelets.

Citation: Chapter 36 Blood Coagulation and Anticoagulant, Fibrinolytic, and Antiplatelet Drugs, Brunton LL, Knollmann BC. *Goodman & Gilman's: The Pharmacological Basis of Therapeutics, 14e;* 2023, figure 36-7. Available at: https://accessmedicine.mhmedical.com/content.aspx?bookid=3191&sectionid=266700467 Accessed: April 11, 2023

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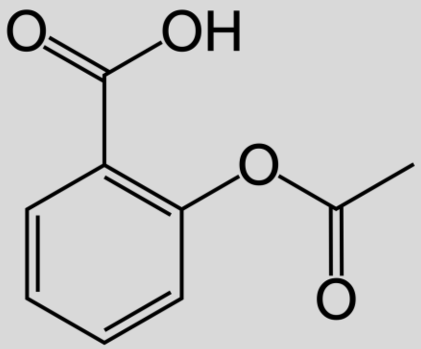
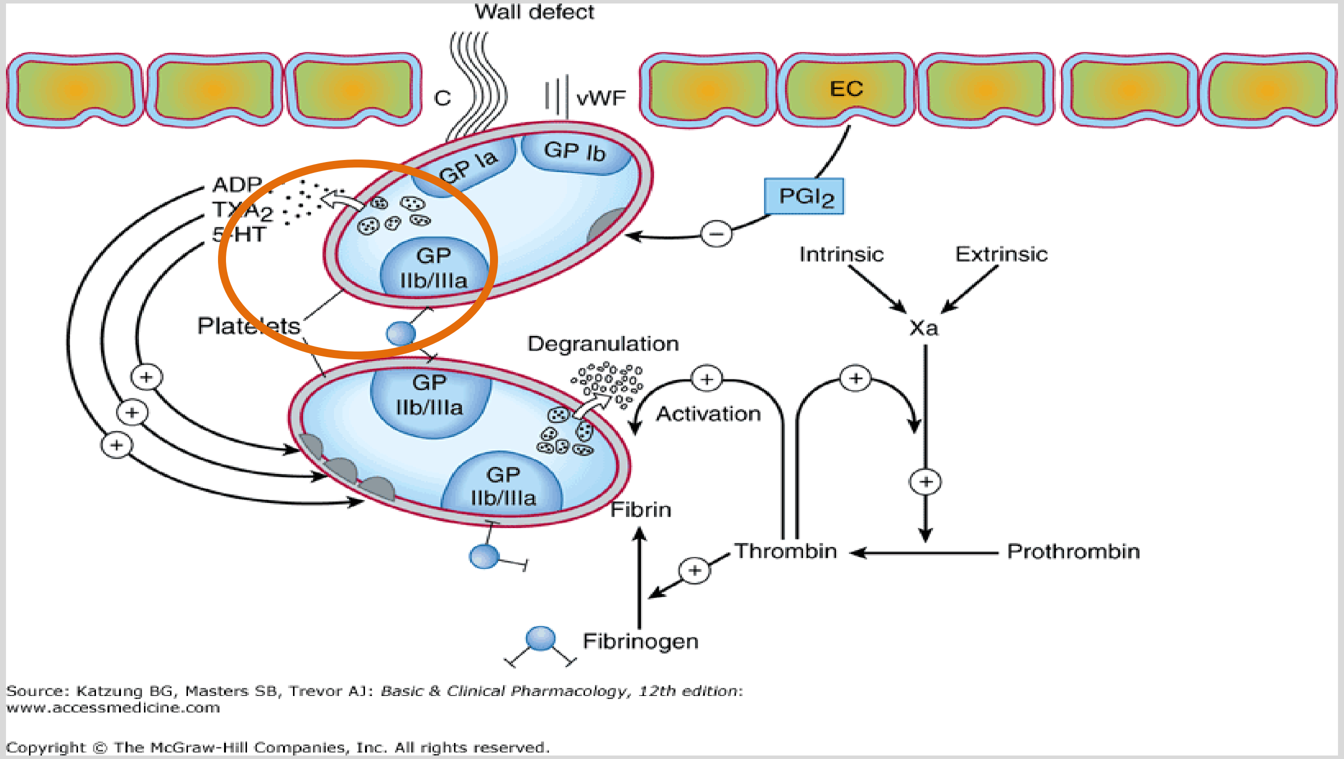
Platelet COX-1 Inhibitor

∗Aspirin



aspirin = acetylsalicylic acid



Thromboxane A_2 (TXA_2) is the major cyclooxygenase (COX-1) product in platelets. TXA_2 has autocrine / paracrine actions as a potent inducer of local platelet aggregation and vasoconstriction. It activates adjacent platelets, amplifying the action of other platelet agonists.

Aspirin Pharmacokinetics and Mechanism of Antiplatelet Action

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Aspirin is taken up by platelets in the portal circulation before it is deacetylated to salicylate on the first pass through the liver.

- Oral
- ASA $\xrightarrow{\text{esterases}}$ salicylate
  - gut mucosa
  - blood
  - red blood cells
  - synovial tissues
- Salicylate: hepatic conjugation
- Renal excretion of
  - salicyluric acid (75%)
  - salicylic acid (10%)
- t½ in portal circulation ~20 min

Mechanism:

Aspirin irreversibly acetylates an allosteric site

↓

Inhibits TXA_2 synthesis

↓

TXA2 actions are prevented

↓

reduces platelet aggregation and vasoconstriction

- Effects lasts for the life of the platelet (7-10 days).
- **Thus, repeated LOW doses of aspirin produce a cumulative effect on platelet function.**

Aspirin's Antiplatelet Uses and Cautions

Dual therapy with P2Y12 receptor inhibitor for some indications.

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Therapeutic Uses:

Cardiovascular protection to reduce the risk of thrombotic events – heart attack and ischemic stroke – in high-risk patients

Secondary prevention:

- To reduce risk of subsequent CV disease events

Primary prevention

- High-risk patients with established CVD
- Intermediate- and high-risk in discussion with patient regarding individual preferences and risk/benefit

Adjunct in revascularization procedures, such as PCI, CABG, stenting

PCI: percutaneous coronary intervention;
CABG: Coronary artery bypass graft

Cautions:

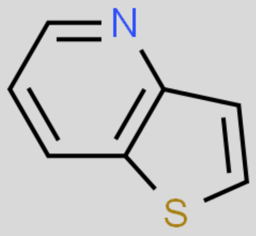
- Upper gastrointestinal bleeding
- Surgery: Avoid use1-2 weeks prior to surgery except if patient has previous cardiac stent placement
- Hypersensitivity: Aspirin-induced urticaria, rhinosinusitis, bronchospasm, anaphylaxis
- Pregnancy: Low doses for certain medical conditions have not been shown to cause fetal harm.

  Recommended to discontinue prior to delivery.
- Children with viral illness: Association with Reye's syndrome

ADP receptor antagonists: $P2Y_{12}$ Receptor Inhibitors

Irreversible inhibitors:

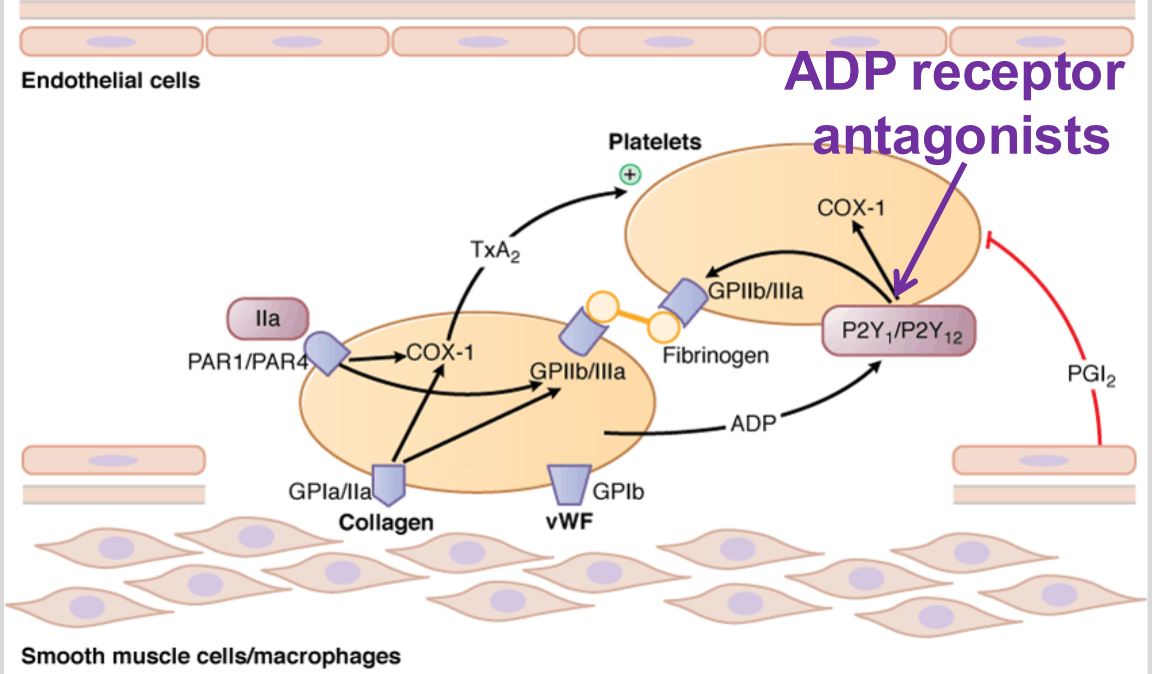
Thienopyridines:



*Clopidogrel
Prasugrel

Reversible inhibitors:

*Ticagrelor
*Cangrelor



Source: Brunton LL, Chabner BA, Knollmann BC: *Goodman & Gilman's The Pharmacological Basis of Therapeutics, 12th Edition*: www.accessmedicine.com

$P2Y_{12}$ is an ADP G_iPCR on platelets. Inhibition of adenylyl cyclase decreases cAMP concentrations and leads to other downstream signaling events resulting in platelet aggregation.

$P2Y_1$ is a platelet ADP GqPCR that activates the PLC–IP_3–Ca^{2+} pathway, which induces an activating conformational change in GP IIb/IIIa receptors, resulting in platelet aggregation.

Both receptor types must be activated by ADP to cause platelet aggregation.

Pharmacokinetics

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	Irreversible Inhibitors		Reversible Inhibitors	
	Clopridogrel	Prasugrel	Ticagelor	Cangrelor*
A	Oral prodrug Onset Day 2	Oral prodrug Onset <30 min	Oral Onset ~2 h	IV bolus f/b infusion T_{max} <2 min
D	Protein binding 98%	98% (active metab)	>99%	98%
M	CYP2C19 conversion to active thiol metab (15% of dose)	Hydrolysis then CYP3A4, 2B6 to active thiol metab	CYP3A4 active thiol metab	Rapid dephosphorylation in plasma, inactive
E	Urine, feces	Urine, feces	Urine, feces	Urine, feces
t½	Clopidogrel ~6 h Active metab ~30 min Platelet bound ~11 dys	Active metab ~7 h	Ticagrelor ~24 h after drug is discontinued	3-6 min Rapid return of platelet function

Activation of prasugrel is more efficient → faster onset of action and greater, more predictable inhibition of ADP-induced platelet aggregation

*Cangrelor is an ADP analog

Therapeutic Uses / Adverse Effects

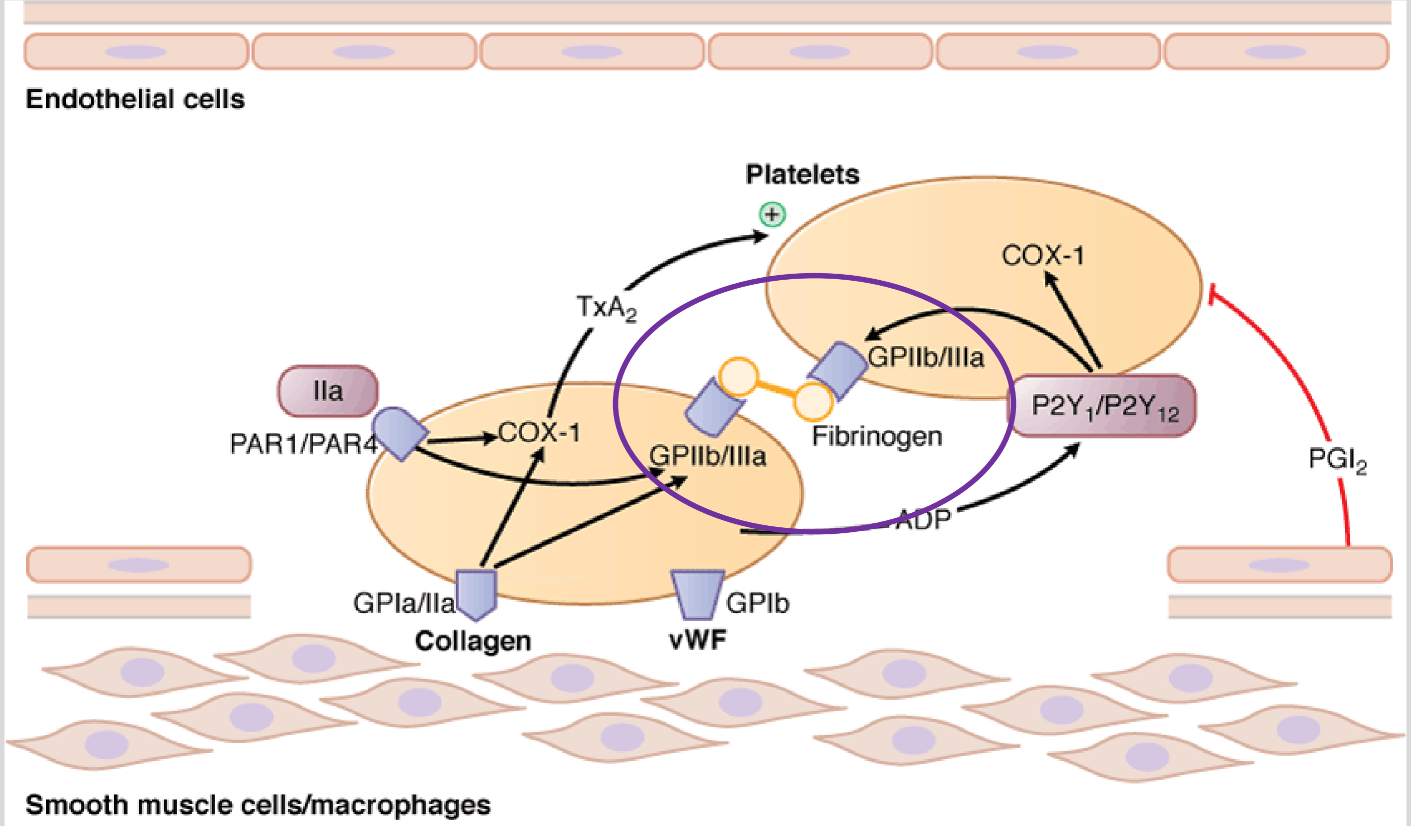
<table>
<tr><th colspan="2">Irreversible Inhibitors</th><th colspan="2">Reversible Inhibitors</th></tr>
<tr><td colspan="2">Clopridogrel and Prasugrel</td><td>Ticagelor</td><td>Cangrelor</td></tr>
<tr><td colspan="2">Dual therapy with aspirin
Secondary CV protection to reduce risk of thrombotic events
• STEMI: following MI with or without reperfusion therapy
• Acute coronary syndrome: After non-STEMI or in unstable angina
• PCI: After bare metal stent or drug-eluting stent placement
• Patients intolerant to aspirin after STEMI</td><td>• Acute coronary syndrome: patients with a history of MI or unstable angina
• Coronary artery disease, stable (primary prevention)
• Minor ischemic stroke</td><td>• Adjunct to PCI to patients who have not been treated with a $P2Y_{12}$ inhibitor or GP IIb/IIIa inhibitor
• Bridging therapy prior to cardiac surgery following thienopyridine discontinuation</td></tr>
<tr><td colspan="4">Adverse effects: All antiplatelet drugs are associated with bleeding.</td></tr>
<tr><td>CYP2C19 poor metabolizers →↓ conversion to active thiol</td><td>Bleeding, higher rates than with clopidogrel
• Not recommended in patients ≥75 years</td><td>• Thrombocytopenic purpura
• Bradyarrhythmia
• Dyspnea (resolves)</td><td>Bleeding</td></tr>
</table>

Glycoprotein IIb/IIIa Receptor Antagonists
✳Eptifibatide
Tirofiban

GP Iib/IIIa is a platelet-surface integrin complex. (~80,000 are present on platelet surface.) They are inactive on resting platelets.

Activation of platelets leads to a conformational change and activation.

GP Iib/IIIa bind fibrinogen, which anchors platelets to each other and to endothelial cells.



Source: Brunton LL, Chabner BA, Knollmann BC: *Goodman & Gilman's The Pharmacological Basis of Therapeutics, 12th Edition:* www.accessmedicine.com

Abciximab is Fab fragments of an immunoglobulin that targets GP IIb/IIIa. It is no longer manufactured and is not included in this presentation.

GP IIb/IIIa Inhibitors: Pharmacology

Eptifibatide	Tirofiban
Cyclic heptapeptide	Synthetic nonpeptide
t½ ~2.5 hours IV bolus	t½ ~2 hours IV bolus
Platelet function restored in ~4 hours	Platelet function restored in 4-8 hours

GP IIIb/IIIa Inhibitors reversibly block platelet aggregation induced by any agonist.

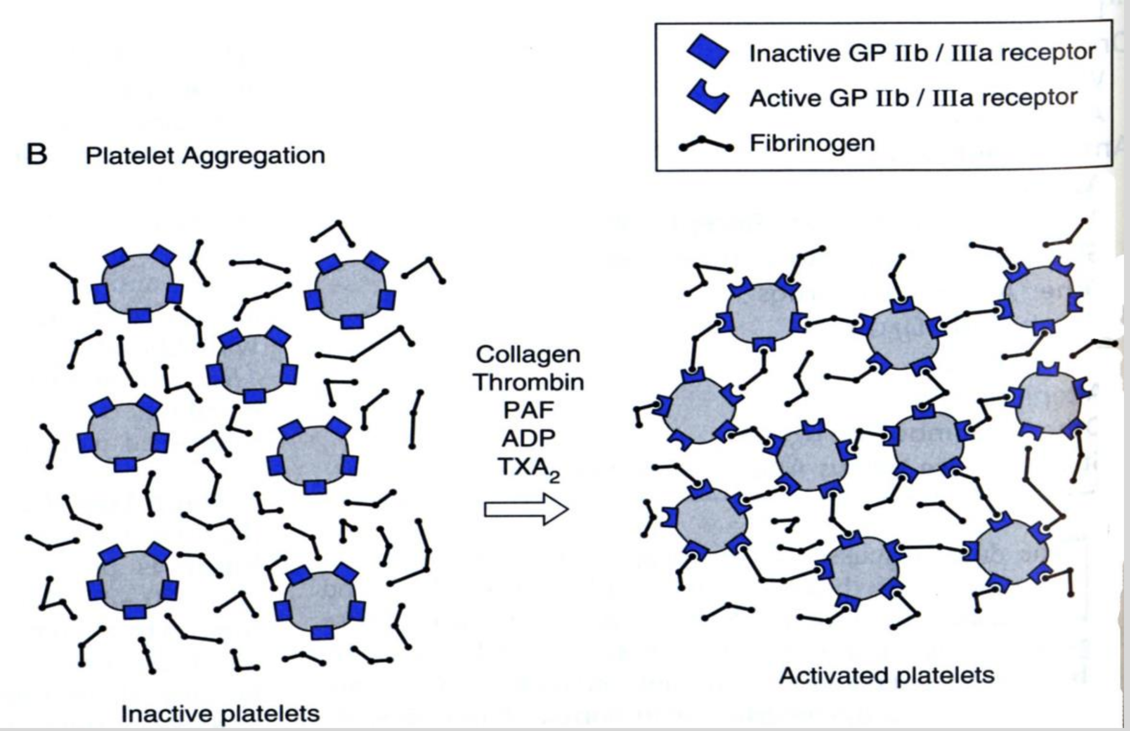
↓

reversibly block fibrinogen binding and bridging of platelets

↓

prevent platelet aggregation

Contraindications	
·Active abnormal bleeding within the previous 30 days ·History of bleeding diathesis ·History of stroke within 30 days ·History of hemorrhagic stroke ·PT >1.2 times control or INR ≥2.0	·Severe hypertension not adequately controlled on medication ·Major surgery within the preceding 6 weeks ·Dependency on hemodialysis ·History of intracranial disease ·Thrombocytopenia ·Hypersensitivity



Antidotes

Protamine sulfate

Vitamin K

Prothrombin Complex concentrate

Idarucizumab

Andexanet alfa (apixaban and rivaroxaban reversal)

Agents for Reversal of Anticoagulant Activity

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Antidote	Anticoagulant	Mechanism
Protamine	Heparin	Highly cationic protein combines with acidic UHF and LMWH forming stable complex that is excreted • Acts in ~5 minutes; t½ ~7 minutes • Protamine has weak anticoagulant effects, therefore excessive protamine dosing may worsen bleeding.
Vitamin K Phytonadione	Warfarin	Outcompetes warfarin at VKORC1 • Oral onset of action 6-10 hours; INR returns to normal in 24-48h; may cause warfarin resistance
Prothrombin complex concentrate (also FFP)	Urgent reversal of VKAs and DOACs	4-factor PCC increases the levels of vitamin K-depended coagulation factors II, VII, IX, and X plus protein C and S • May predispose the patient to a thromboembolic complication (weigh risk *v* benefit)
Idarucizumab	Oral direct thrombin inhibitor	MAb fragment that specifically binds dabigatran and its active acyglucuronide metabolites • Onset in minutes; duration 24h or more

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Summary

Thrombolytics and Antiplatelet Agents

- Thrombolytics
  - Plasminogen activators
    - tPA: Alteplase, Tenecteplase, Reteplase
    - Streptokinase
    - Urokinase

Treatment of arterial and venous thrombosis

- Antiplatelet agents
  - COX inhibitor
    - Aspirin
  - Glycoprotein IIb/IIIa inhibitors
    - Abciximab
    - Eptifibatide
    - Tirofiban
  - ADP receptor inhibitors ($P2Y_{12}$ component)
    - Clopridogrel
    - Prasugrel
    - Ticagrelor
    - Cangrelor

Treatment and prevention of occlusive cardiovascular diseases and maintenance of vascular grafts and arterial patency

SUMMARY OF FIBRINOLYTIC AGENTS

Streptokinase (n/a in U.S.)	• 47,000 Da protein produced by β-hemolytic streptococci • It has no intrinsic enzymatic activity • 1:1 complex with plasminogen → conformational change → exposes plasminogen's active site → complex cleaves circulating (free) plasminogen → forms plasmin • Activates both free and fibrin-bound plasminogen ⇒ lytic state • Cross reactivity with circulating anti-streptococcal antibodies → may cause allergic reactions or neutralize the activity of streptokinase
Urokinase	• Human enzyme synthesized by the kidney • Directly converts plasminogen to plasmin • Activates both free and fibrin-bound plasminogen
Alteplase (t½ 5 min)	• Recombinant t-PA • Preferentially activates plasminogen bound to fibrin
Reteplase (t½ 13-16 min)	• Recombinant t-PA variant; lacks the finger, EGF, and first kringle domains → more weakly binding of fibrin; longer t½ (2 bolus doses)
Tenecteplase (t½ 90-130 min)	• Recombinant t-PA variant; longer t½ (1 bolus dose) • Relatively resistant to plasminogen activator inhibitor-1 compared to alteplase and reteplase

Summary of Antiplatelet Agents

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- Aspirin acetylates cyclooxygenase-1 (COX-1) and inhibiting TXA2 synthesis. Low-dose aspirin effectively inhibiting platelet aggregation for 5 to 7 days (the life of the platelet).
- ADP receptor P2Y12 inhibitors prevent activation of the GP IIb/IIIa receptor complex, thus preventing fibrinogen cross-linking. Clopidogrel and prasugrel (thienopyridines) are irreversible inhibitors. Reversible inhibitors are ticagrelor (oral) and cangrelor (I.V.)
- GP IIb/IIIa inhibitors abciximab, eptifibatide, and tirofiban block platelet aggregation by blocking the binding of fibrinogen, thereby preventing fibrinogen cross-linking.

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